AdventureQuestEngine Game Manual

Overview

AdventureQuestEngine is a sophisticated, terminal-based C++ adventure game engine. It supports a myriad of deep RPG mechanics, including character creation, a heavily tactical turn-based combat system, extensive magic, crafting, procedural exploration, and dynamic economy.

The Game Engine & Architecture

The GameEngine class serves as the central hub coordinator. The core loop handles exploration, combat transitions, menu navigation, and character status tracking. The system features:

- Character Creation: Selection of classes, name, and application of base stats suited to different playstyles.
- Exploration & UI: Grid-based movement through rooms with fog of war, hidden items detection, and rich contextual menus.
- Save & Load (File I/O): Progress tracked in local save files, along with robust data mechanisms like leaderboards.

Advanced Combat System

Combat is highly tactical, combining positioning, combos, and elemental affinities.

- Positioning Mechanics: Characters choose between Front Line (high damage and risk), Mid Line (balanced), and Back Line (safe, ranged/magic only).
- Diverse Actions: Basic/Heavy/Quick attacks, Defend (+50% defense, +30% dodge), Counterattacks, fleeing, casting spells, and using items.
- Combo Chains: Successful attacks build combo meters capable of triggering powerful Combo Finishers.
- Elemental System: Physical, Fire, Ice, Lightning damage types interacting with specific elemental resistances/vulnerabilities.
- Detailed Outcomes: Attacks can critically hit, be fully blocked, dodged, or "graze" (deal 50% damage) while inflicting various status effects.

Magic System & Spells

Character classes with access to magic can learn spells across a variety of mystical schools:

- Evocation: Firebolt, Frostbolt, Chain Lightning, Inferno, Blizzard.
- Abjuration: Protective spells like Magic Shield and Reflection Aura.
- Transmutation: Buffs and debuffs like Strength Boost, Enfeeblement, Haste, and Slowness.
- Necromancy: Life drain, undead summoning, and curses.
- Restoration: Heal, Greater Heal, and Dispel Magic for cleansing status effects.
- Conjuration & Divination: Arcane Missiles, Meteor Shower, and utility spells like Analyze Weakness.

Crafting, Economy & Merchants

A full-fledged economy and crafting progression is integrated into the engine.

- Materials: Wood, Metal, Leather, Cloth, Gem, Essence, Herb, Bone, Dragon Scale, Crystal.
- Recipe System: Discoverable recipes requiring specific materials, gold, level, and crafting stations (e.g., Forge, Alchemy Lab).
- Enchantment System: Upgrades applied to items up to +10. Includes elements like Lifesteal, Thorns, Regeneration, Unbreaking, and Luck.
- Item Durability: Equipment degrades with use and requires management.
- Merchants: Various vendors including General Stores, Weaponsmiths, Black Markets, and Traveling merchants that dynamically restock. Reputation impacts buy/sell prices.
- Bartering: System allows exchanging specific items for merchant wares with complex gold differentials.

Additional Features

To flesh out the RPG experience, the engine supports:

- Companions: AI partners that fight alongside the player and level up.
- EventsSystem: Dynamic occurrences during exploration that trigger unique challenges.
- Skill Trees: Branching paths for progression as characters level up.
- Achievements: Trackable milestones and statistics for completionists.